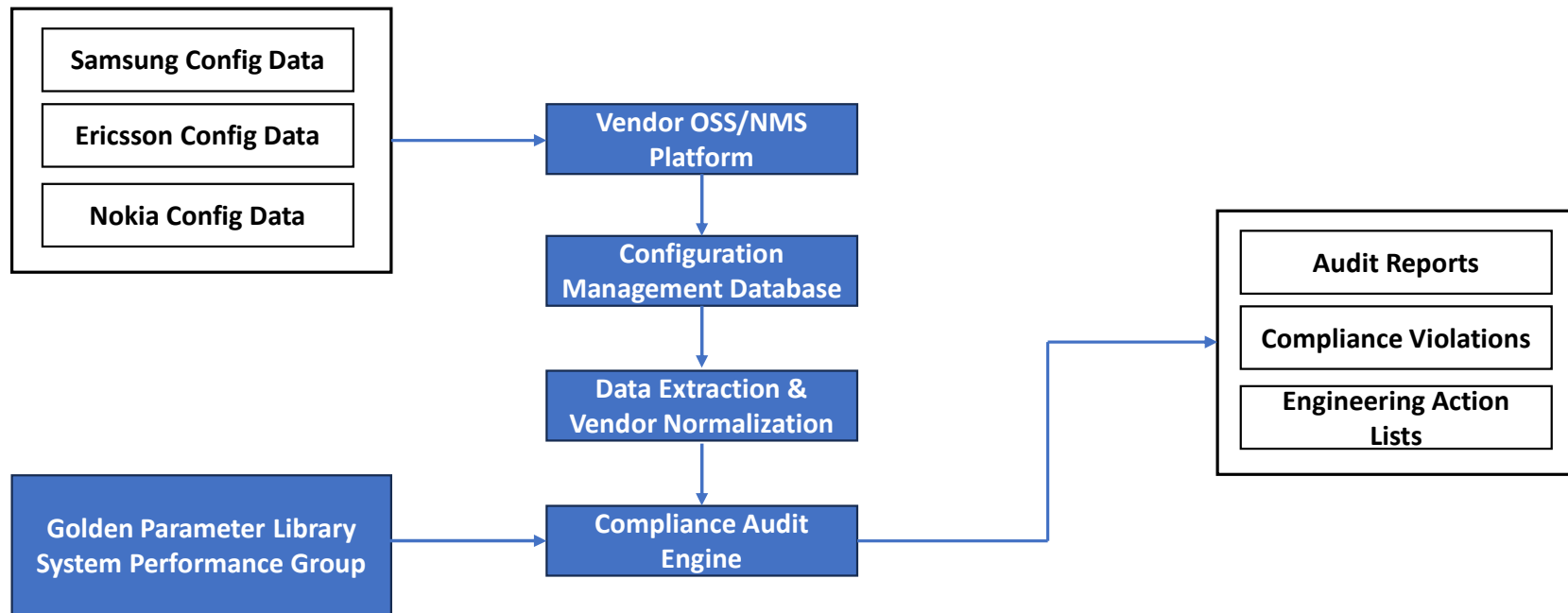


Wireless Priority Services Compliance Analytics Platform

Automated Multi-Vendor Configuration Compliance Auditing Supporting 125,000+ Network Elements

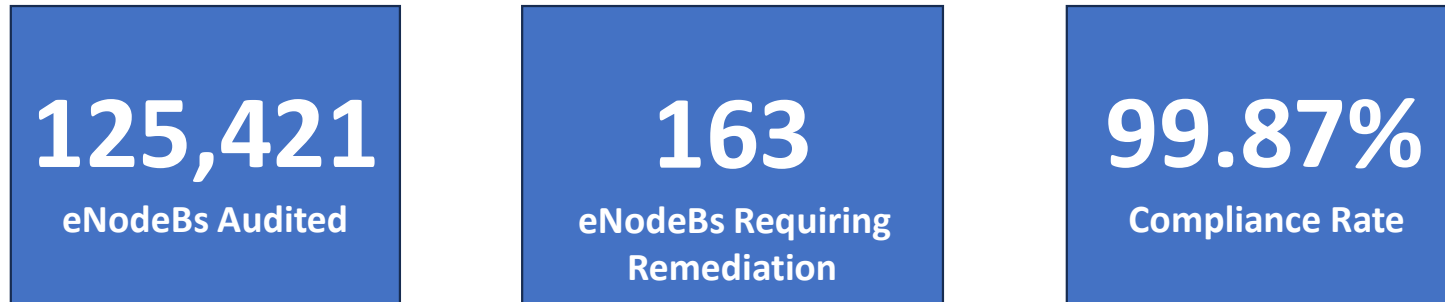


Golden Parameter Validation Example

Network Elements	Parameter	Required Value	Actual Value	Status
Site-001_Hospital	Priority Access Feature	Enabled	Enabled	PASS
Site-028_Suburban	Pooling Capability	Pooling Enabled	Pooling Disabled	FAIL
Site-036_Business	Maximum Call Capacity	600	400	FAIL

Enterprise Compliance Audit Summary

Automated Validation of Wireless Priority Services Configuration Standards



Key Outcomes

- Automated compliance auditing across 125,421 eNodeBs.
- Identified 163 non-compliant eNodeBs requiring engineering remediation.
- Achieved 99.87% configuration compliance for Wireless Priority Services.

Project Summary

CHALLENGE

- Priority Services required consistent configuration across thousands of LTE network elements.
- Configuration standards differed across Samsung, Ericsson, and Nokia networks.
- Manual compliance audits were time-consuming, difficult to scale, and prone to inconsistencies.
- Non-compliant configurations could impact Priority Services functionality and contractual obligations.

SOLUTION

- Developed a multi-vendor compliance analytics platform using Python.
- Extracted and normalized configuration data from Samsung, Ericsson, and Nokia networks.
- Validated network configurations against Golden Parameter standards.
- Automated pass/fail compliance auditing and exception identification.
- Generated compliance reports and engineering remediation lists.

RESULTS

- Automated compliance auditing across 125,421 LTE network elements.
- Identified 163 non-compliant eNodeBs requiring engineering remediation.
- Achieved 99.87% configuration compliance for Wireless Priority Services.
- Reduced manual audit effort and improved consistency.
- Enabled scalable auditing across multi-vendor LTE networks.

Skills Demonstrated

Data Engineering

Analytics & Governance

Tools & Technologies

- | | | |
|---------------------------------|-----------------------------------|---------------------------|
| • Multi-vendor Data Integration | • Compliance Analytics | • Python |
| • Data Extraction | • Rule-Based Validation | • Pandas |
| • Data Normalization | • Exception Detection & Reporting | • Excel |
| • Data Validation | • KPI Reporting | • Configuration Databases |
| • Configuration Data Management | • Audit Traceability | • Automated Reporting |